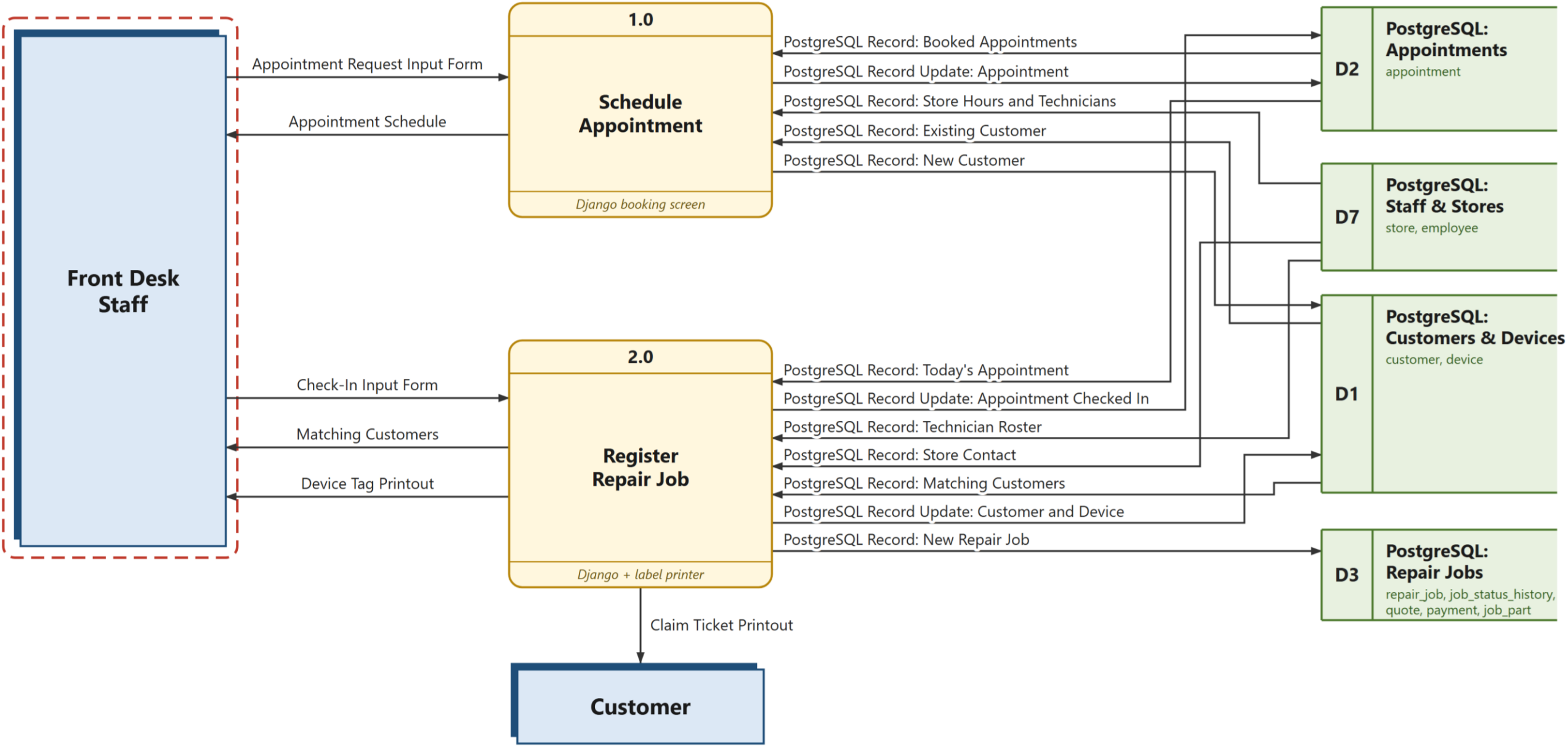
Sprint 3 Presentation

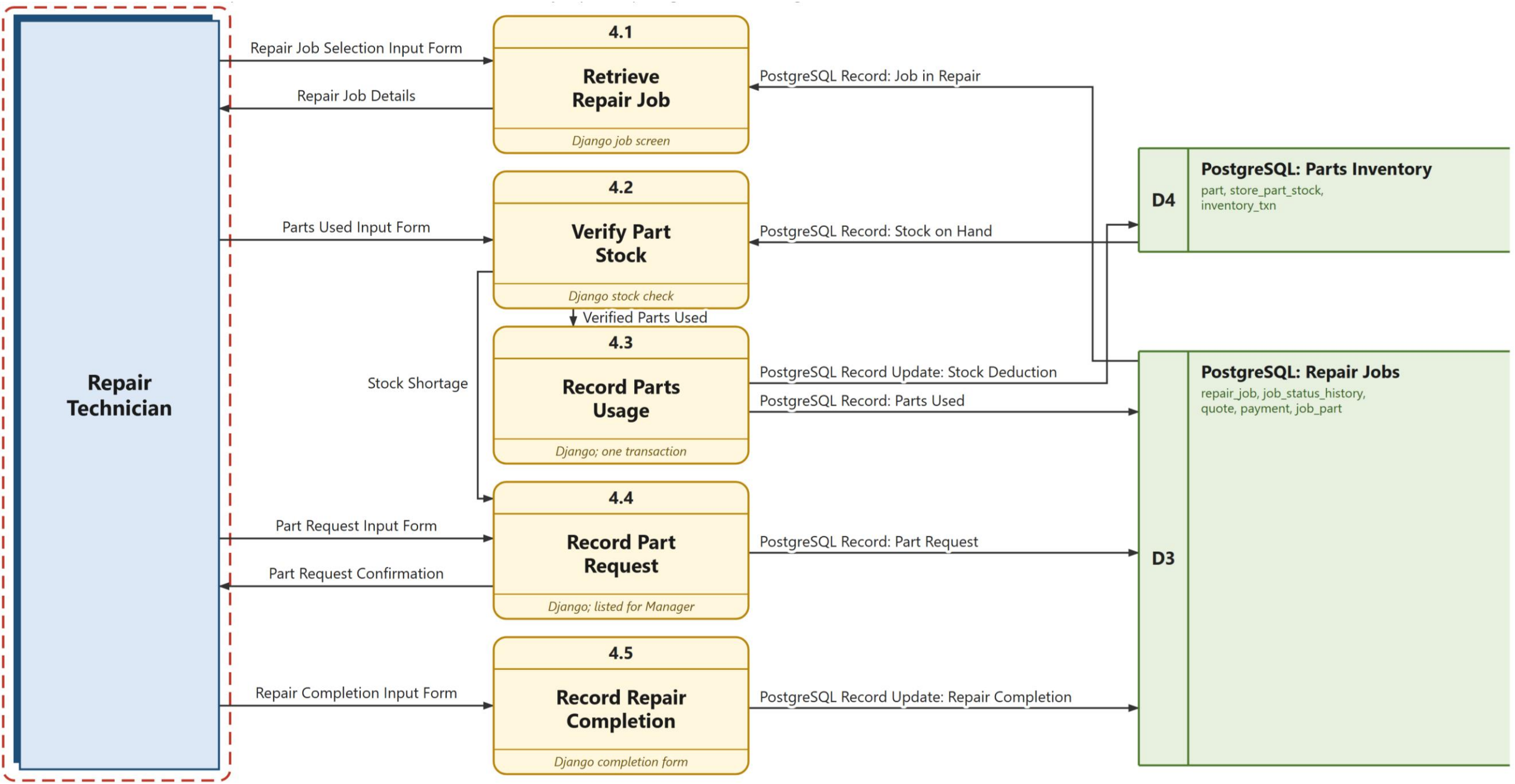
What changed since Sprint 2

- Modified logical DFDs naming to reflect what the system does (Record Diagnosis, Record Repair Work, etc.)
- New use cases to for appointments, and view repair status
- ERD simplified from 22 to 16 entities
- DFDs and ERD aligned: 7 data stores hold the 16 entities, each entity in one data store.
- Built physical DFDs & EFDs
- Built three new prototypes

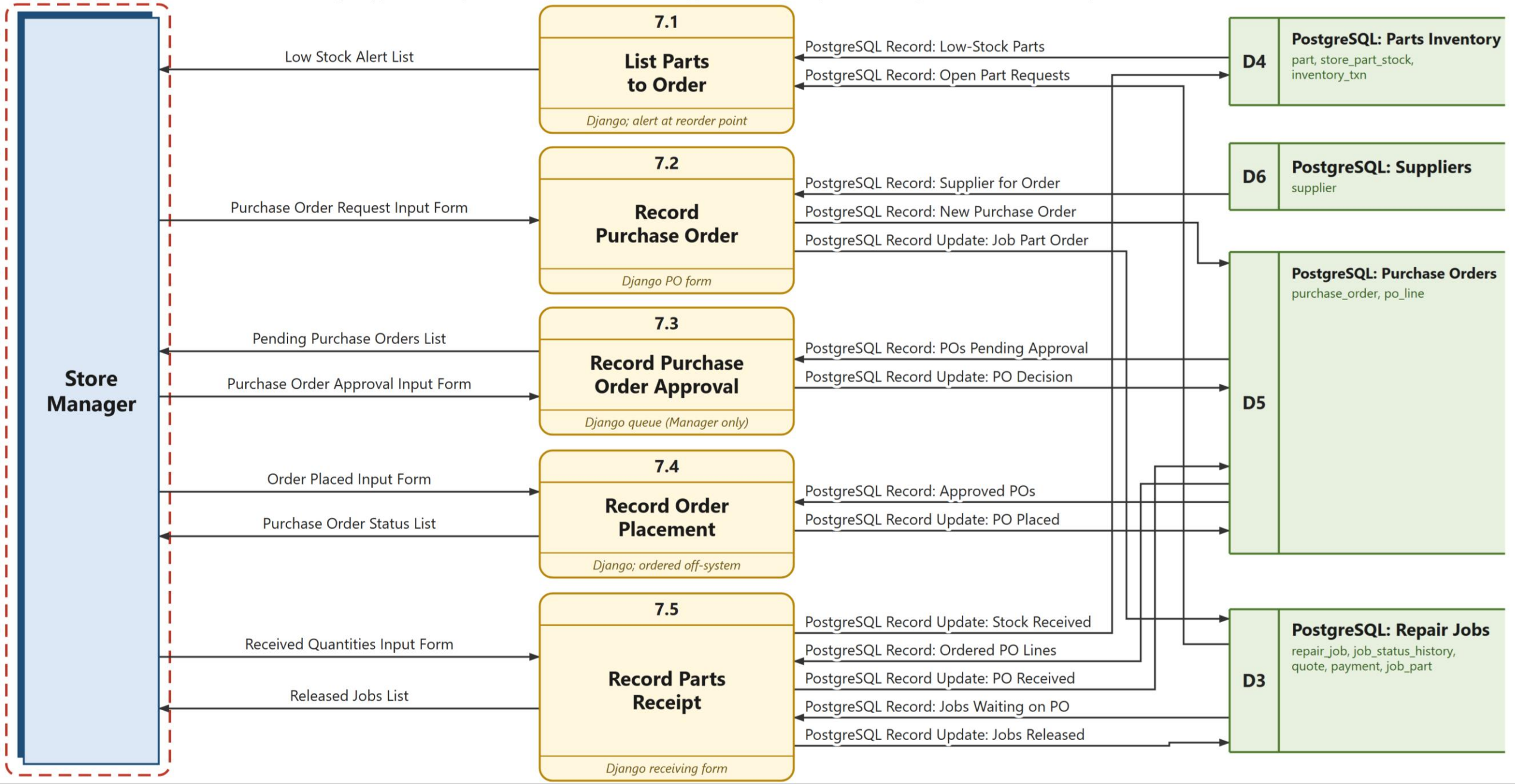
Physical DFD for Appointments & Check In (Level 0)



Physical DFD for Device Repair (Level 1)

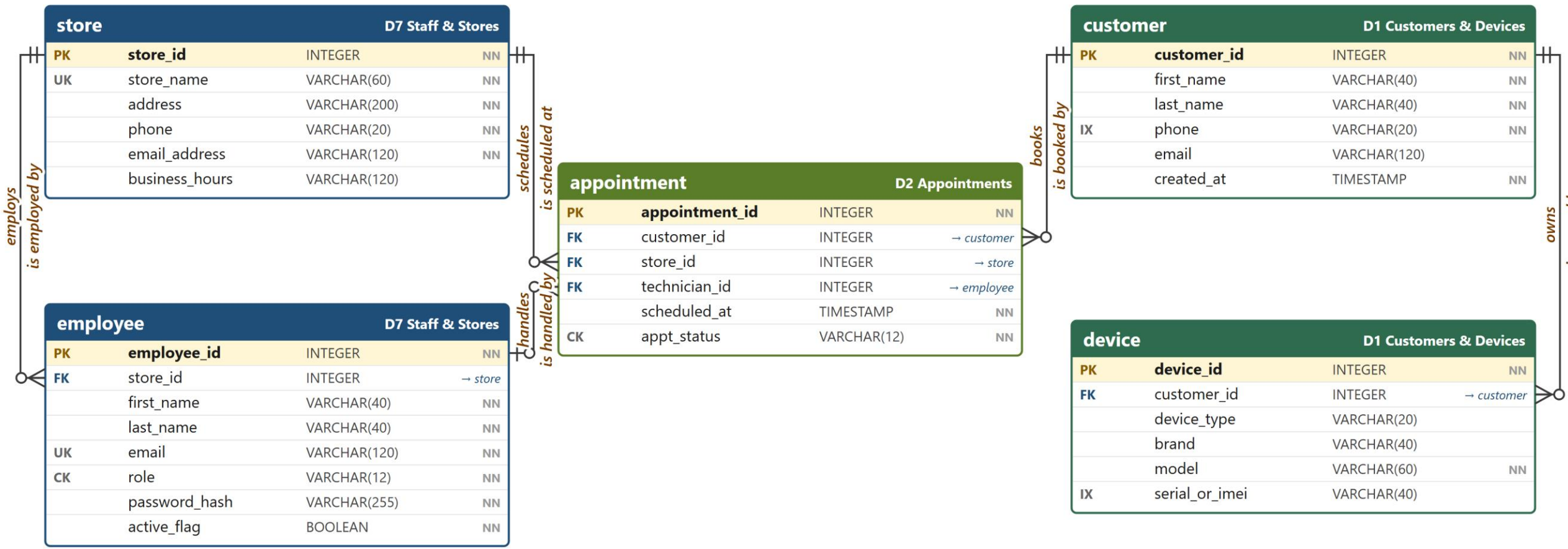


Physical DFD for Purchase Orders (Level 1)



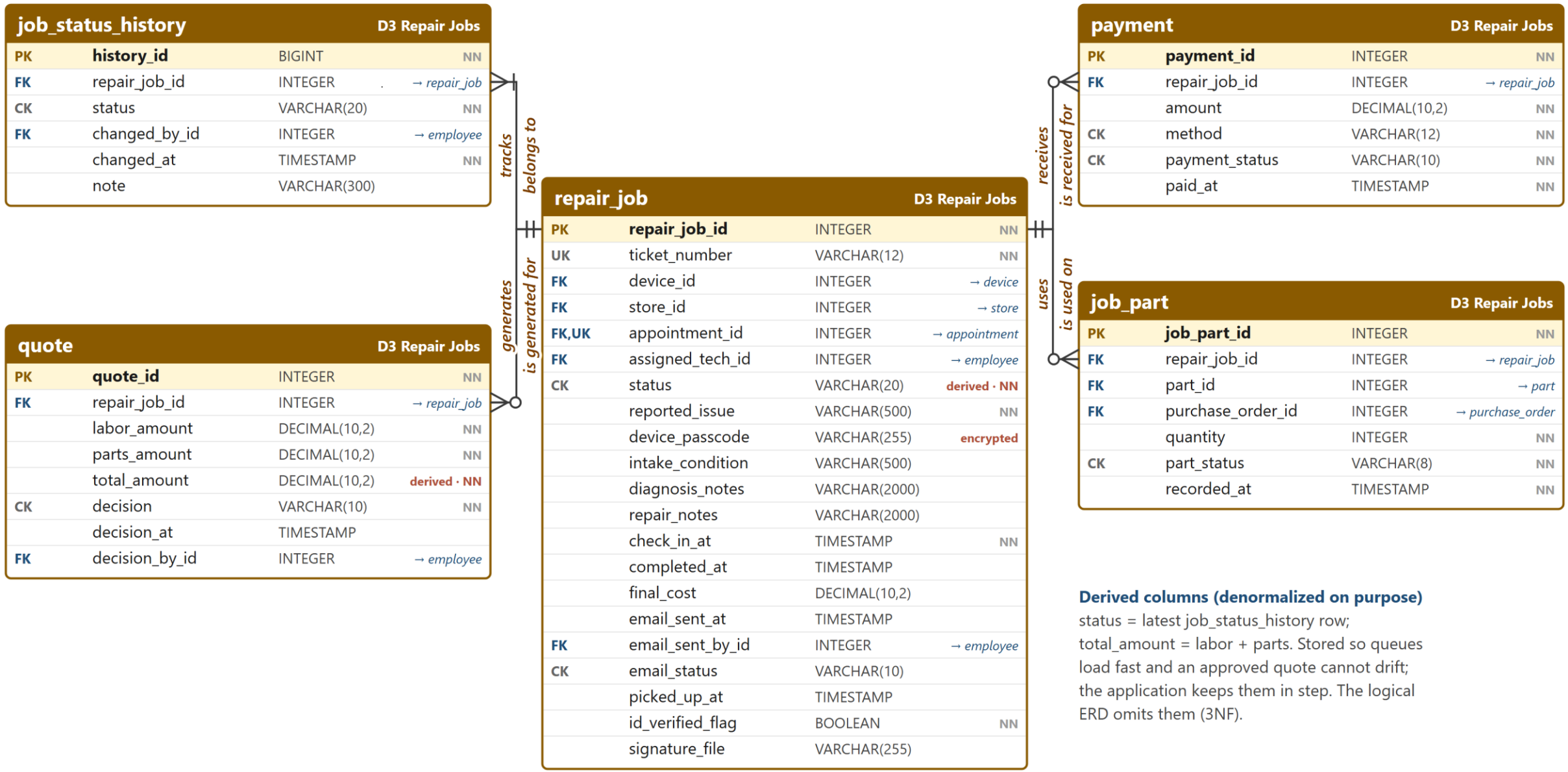
QuickPhone — Physical ERD detail 1 of 3: Customers, devices, appointments, staff and stores

D1 Customers & Devices · D2 Appointments · D7 Staff & Stores. PK primary key · FK foreign key (→ table) · UK unique · CK checked value list · NN not null



QuickPhone — Physical ERD detail 2 of 3: Repair jobs

D3 Repair Jobs: the job, its status history, quotes, payment and parts. PK primary key · FK foreign key (→ table) · UK unique · CK checked value list · NN not null

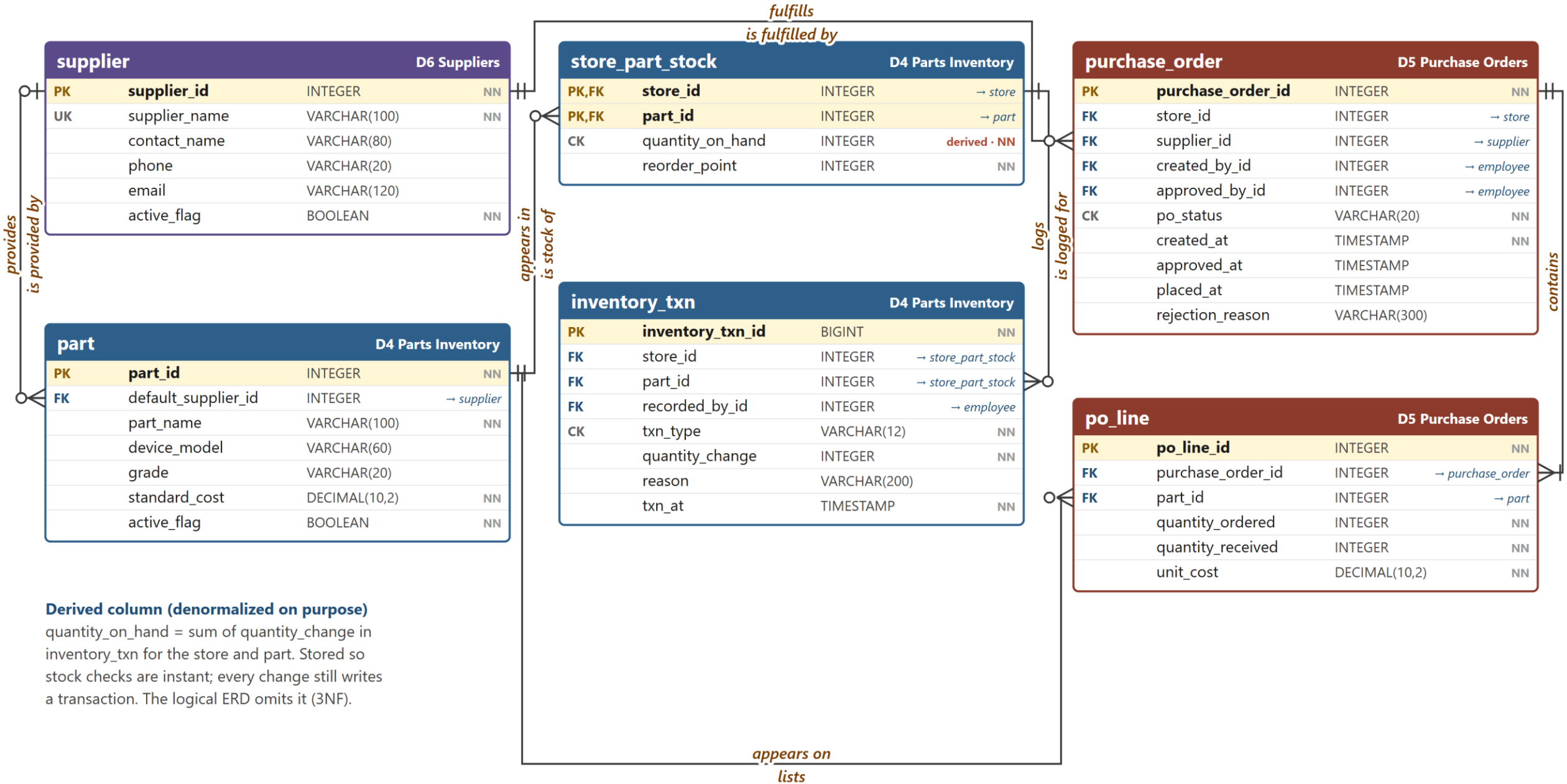


Derived columns (denormalized on purpose)

status = latest job_status_history row; total_amount = labor + parts. Stored so queues load fast and an approved quote cannot drift; the application keeps them in step. The logical ERD omits them (3NF).

QuickPhone — Physical ERD detail 3 of 3: Parts, purchasing and suppliers

D4 Parts Inventory · D5 Purchase Orders · D6 Suppliers. PK primary key · FK foreign key (→ table) · UK unique · CK checked value list · NN not null



supplier — D6 Suppliers

PK	**supplier_id**	INTEGER	NN
UK	supplier_name	VARCHAR(100)	NN
	contact_name	VARCHAR(80)	
	phone	VARCHAR(20)	
	email	VARCHAR(120)	
	active_flag	BOOLEAN	NN

store_part_stock — D4 Parts Inventory

PK,FK	**store_id**	INTEGER	→ store
PK,FK	**part_id**	INTEGER	→ part
CK	quantity_on_hand	INTEGER	derived · NN
	reorder_point	INTEGER	NN

purchase_order — D5 Purchase Orders

PK	**purchase_order_id**	INTEGER	NN
FK	store_id	INTEGER	→ store
FK	supplier_id	INTEGER	→ supplier
FK	created_by_id	INTEGER	→ employee
FK	approved_by_id	INTEGER	→ employee
CK	po_status	VARCHAR(20)	NN
	created_at	TIMESTAMP	NN
	approved_at	TIMESTAMP	
	placed_at	TIMESTAMP	
	rejection_reason	VARCHAR(300)	

part — D4 Parts Inventory

PK	**part_id**	INTEGER	NN
FK	default_supplier_id	INTEGER	→ supplier
	part_name	VARCHAR(100)	NN
	device_model	VARCHAR(60)	
	grade	VARCHAR(20)	
	standard_cost	DECIMAL(10,2)	NN
	active_flag	BOOLEAN	NN

inventory_txn — D4 Parts Inventory

PK	**inventory_txn_id**	BIGINT	NN
FK	store_id	INTEGER	→ store_part_stock
FK	part_id	INTEGER	→ store_part_stock
FK	recorded_by_id	INTEGER	→ employee
CK	txn_type	VARCHAR(12)	NN
	quantity_change	INTEGER	NN
	reason	VARCHAR(200)	
	txn_at	TIMESTAMP	NN

po_line — D5 Purchase Orders

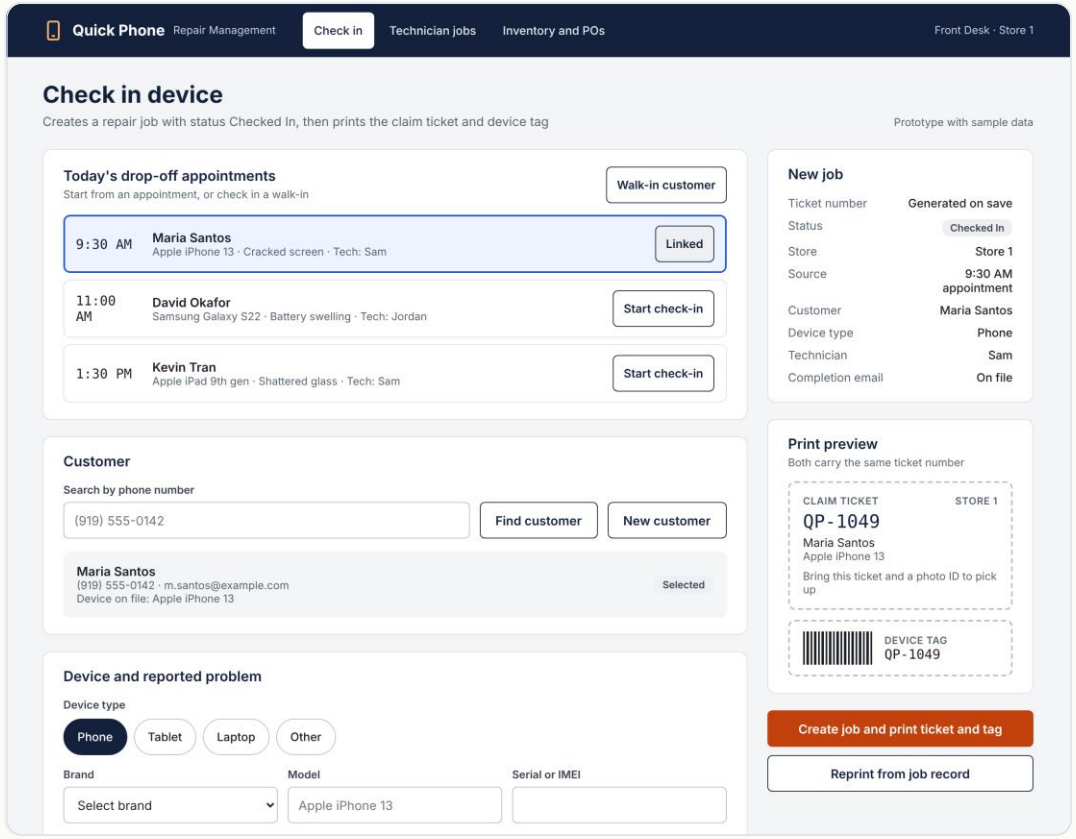
PK	**po_line_id**	INTEGER	NN
FK	purchase_order_id	INTEGER	→ purchase_order
FK	part_id	INTEGER	→ part
	quantity_ordered	INTEGER	NN
	quantity_received	INTEGER	NN
	unit_cost	DECIMAL(10,2)	NN

Derived column (denormalized on purpose)

quantity_on_hand = sum of quantity_change in inventory_txn for the store and part. Stored so stock checks are instant; every change still writes a transaction. The logical ERD omits it (3NF).

INTERFACE 1 · FRONT DESK

Check In Device



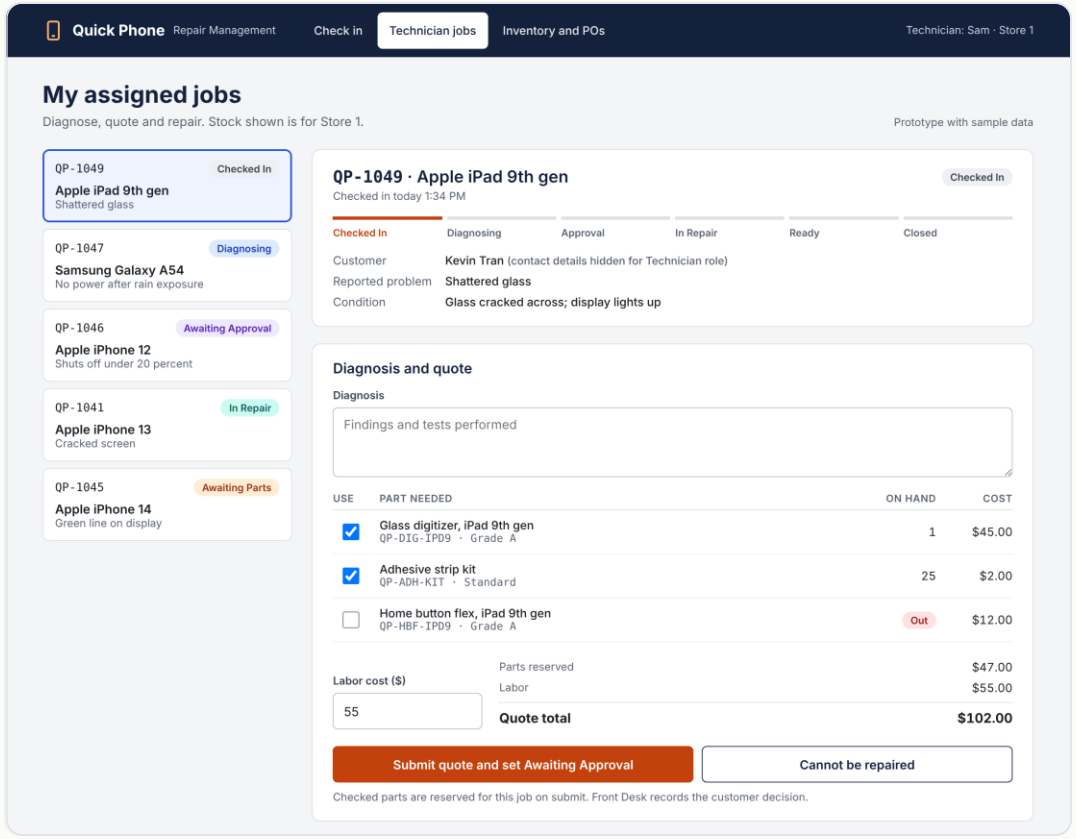
- Starts from a booked appointment or a walk-in, then searches by phone so a returning customer is not entered twice
- Records device, serial or IMEI, condition and reported problem, then assigns a technician by open workload
- Creates the job as Checked In and prints a matching claim ticket and device tag
- Flags a customer with no email so staff call at completion (alternate flow 2a)

TRACES TO

Use case	**UC-01 Check In Device**
DFD process	**2.0 Register Repair Job**
Requirements	**FR02 to FR04, NFR03**
Data stores	**D1, D2, D3, D7**

INTERFACE 2 · TECHNICIAN

Diagnose, Quote and Repair



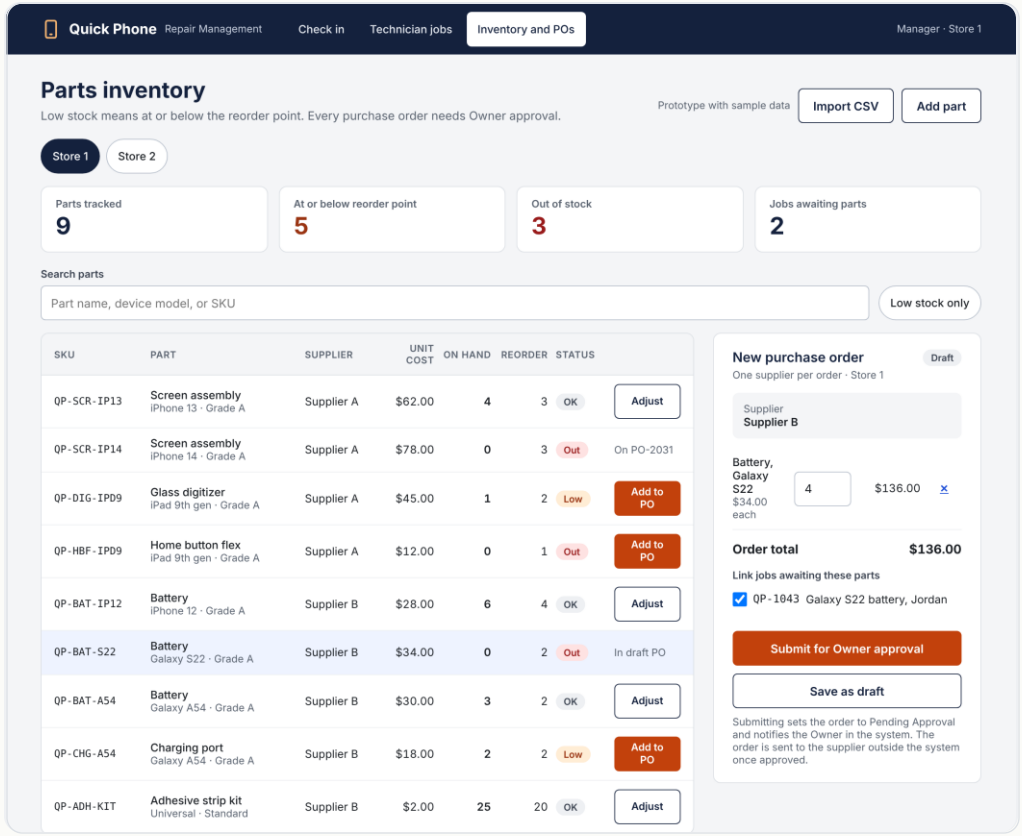
- Shows only jobs assigned to the signed-in technician, using the confirmed job statuses
- Technician picks parts against live store stock and the quote totals automatically
- An out-of-stock part moves the job to Awaiting Parts and adds it to the low-stock list (alternate flow 3a)
- Repair view records parts used and deducts stock; customer contact and email are hidden from this role

TRACES TO

Use cases	**UC-02, UC-04**
DFD process	**3.0 Record Diagnosis, 4.0 Record Repair Work**
Requirements	**FR04 to FR07, NFR01, NFR02, NFR04**
Data stores	**D3, D4**

INTERFACE 3 · MANAGER

Parts Inventory and Purchase Orders



- Stock for each store, with low stock defined as at or below the reorder point
- Add to PO builds a one-supplier draft and links the jobs waiting on those parts
- Every purchase order goes to the Owner for approval before it can be received
- Receiving updates stock, supports partial delivery and returns linked jobs to In Repair

TRACES TO

Use cases	**UC-07, UC-08, UC-10**
DFD process	**7.0 Process Purchase Orders, 9.0 Maintain Shop Records**
Requirements	**FR09 to FR11, NFR01, NFR02**
Data stores	**D3, D4, D5, D6**

Pending Items and Next Steps

- Owners Oct 23 feedback needs to be integrated
- Django + PostgreSQL are provisional
- Sprint 4: build, specifications, and test cases

Thank you